

Lab assignment 2

Lab Ass. 2: On the effect of rotation-based self-supervised learning

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1 Introduction

Self-supervised learning has emerged as a powerful paradigm to exploit large amounts of unlabeled data. In this assignment, we study a rotation-based pretext task and its impact on downstream image classification. Most of our experiments rely on a ResNet-14 convolutional neural network, although in extended trials we also evaluate a deeper ResNet-18 variant to analyse the effect of model capacity. We first train the network to predict image rotations, and then reuse the learned weights to initialize a classifier. The goal is to analyse how self-supervised pretraining affects convergence speed, final performance, and feature representation quality, compared to training a network from scratch.

2 Methodology

Our experiments follow a systematic evaluation protocol built around two tasks: (i) a self-supervised pretext task based on rotation prediction, and (ii) a downstream supervised classification task. Unless otherwise stated, all experiments use the default dataset composed of **15 classes with 480 samples each**. Only one specific experiment—the full-data configuration—uses the extended version of the dataset with **50 classes $\times$ 480 samples**.

The backbone network is a ResNet-14 CNN (see Figure 1 and [He et al. \(2016\)](#)), although a deeper ResNet-18 is also tested to analyse the effect of model capacity. For the pretext task, the network is trained to classify the rotation applied to each image among four possible angles (0° , 90° , 180° , 270°). The same encoder is then reused for the supervised classification task, either initialized with the pretext weights or trained from scratch.

Before conducting full experiments, we run short 10-epoch *sanity-check* tests for both tasks to ensure that the implementation, rotation-label generation, gradients, and optimization behave correctly.

The main experiments are trained for 45 epochs and include the following controlled variations:

- **Baseline configuration (15 $\times$ 480):** ResNet-14, default learning rate and weight decay, and no data augmentation beyond the rotations inherent to the pretext task.
- **Full-data configuration (50 $\times$ 480):** identical to the baseline but using the extended dataset to study the effect of increased class diversity. This is the only experiment that does not use the 15-class dataset.

- **Learning-rate variation:** the initial learning rate is modified (e.g. set to 0.01) to analyse its impact on convergence stability and generalization.
- **Model depth variation:** ResNet-14 is replaced with ResNet-18 to assess whether deeper architectures improve self-supervised representation learning.
- **Data augmentation:** both ResNet-14 and ResNet-18 are trained with stronger geometric and appearance augmentations. For the pretext task we apply padding, random cropping to 64×64 , and random horizontal flips. For classification, we additionally use **ColorJitter**, followed by conversion to tensor and normalization using dataset statistics.
- **Combined configuration:** a final experiment combining several improvements (ResNet-18, reduced learning rate 0.005, weight decay $5 \cdot 10^{-4}$, and full augmentation pipeline).

For each configuration, we track training and validation loss and accuracy for both tasks. We also extract penultimate-layer embeddings to generate t-SNE visualizations for three models: a pretext-only encoder, a fine-tuned classifier (self-supervised initialization), and a classifier trained from scratch.

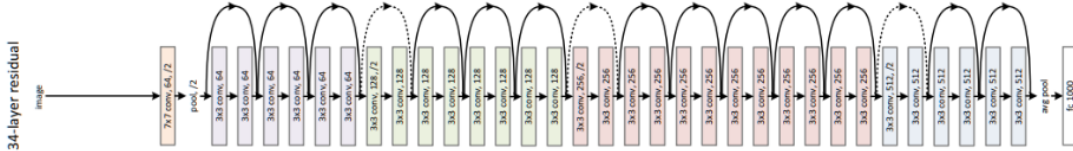


Figure 1: Example of a ResNet-34 architecture from the original paper. Our experiments use a shallower ResNet-14 variant (and, in some trials, a ResNet-18), which follow the same residual design but differ in depth.

3 Training the Pretext Task

3.1 Dataset Inspection

Before training the rotation prediction network, we inspect the pretext dataset to verify that the four rotated variants of each image are generated correctly. Figure 2 shows an example sample under the four rotation angles. These transformations constitute the supervision signal the model must learn to predict. While the semantic content remains unchanged, the geometric orientation differs significantly, providing a strong self-supervised signal that encourages the model to learn orientation-sensitive visual features.

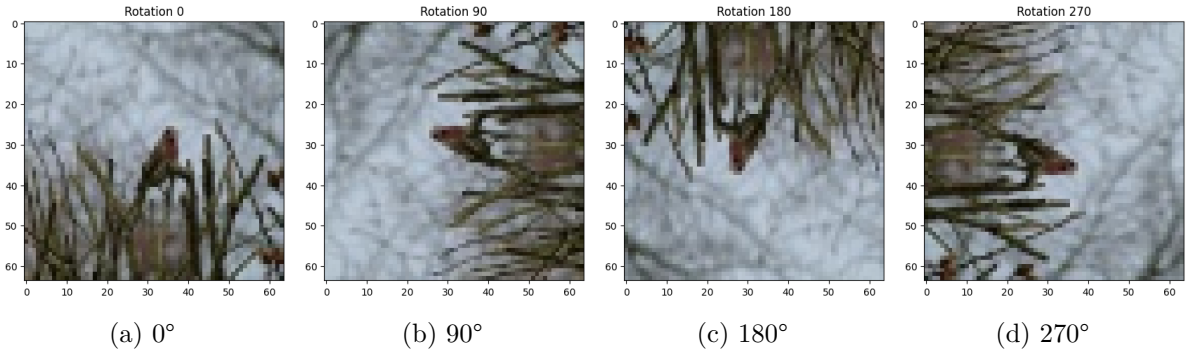


Figure 2: Examples of the four rotated versions of the same training image used in the pretext rotation prediction task.

3.2 Representative Results

Figure 3 reports training and validation loss and accuracy for the baseline pretext configuration (ResNet-14, 45 epochs, 15×480 dataset). These curves summarize the typical behaviour observed across most configurations before introducing stronger regularization and augmentation.

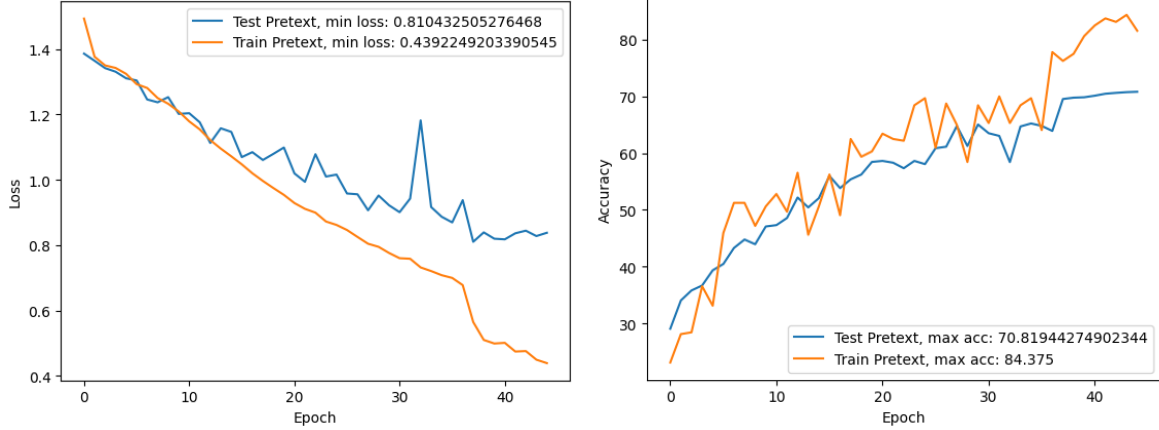


Figure 3: Training and validation loss (left) and accuracy (right) for the baseline pretext task using ResNet-14 trained for 45 epochs on the 15×480 dataset.

The complete set of pretext results, including all epochs, architectures, learning rates, dataset sizes, augmentation settings, and combined configurations, is provided in Appendix A.

3.3 Discussion

Across all configurations, the rotation pretext task is easy for the model to learn: training loss decreases smoothly and the network quickly adapts to the four rotation classes. However, because most experiments use only the reduced 15×480 dataset, the model has limited visual diversity to learn from. This is why the baseline setup shows clear overfitting (Figure 3): training accuracy keeps rising while validation accuracy quickly levels off and becomes unstable.

The experiment using the full 50-class dataset (Appendix A, Fig. 8) confirms this effect. With more classes, validation curves become smoother and the generalization gap shrinks. This suggests that many of the stability issues seen in the smaller dataset would naturally diminish if all experiments used the full data.

Even with the smaller dataset, regularization helps. In particular, data augmentation has the strongest impact. Techniques such as padding, random cropping, and ColorJitter introduce enough variety to noticeably reduce overfitting (Appendix A, Figs. 11, 12). Architectural changes, such as switching from ResNet-14 to ResNet-18, produce smaller improvements unless combined with good hyperparameters and augmentation.

Finally, higher pretext-task accuracy does not necessarily translate into better downstream performance. What matters is the quality and diversity of the representations, not how close the model gets to 100.

Overall, the pretext experiments highlight that data diversity and augmentation play a much larger role than model depth or raw pretext accuracy.

4 Training the Classification Task

4.1 Representative Results: Scratch vs. Self-Supervised Initialization

For the supervised classification task, we compare two initialization strategies: training from scratch with random weights, and fine-tuning from self-supervised pretext weights. Figure 4 shows the representative curves for the baseline configuration over 45 epochs.

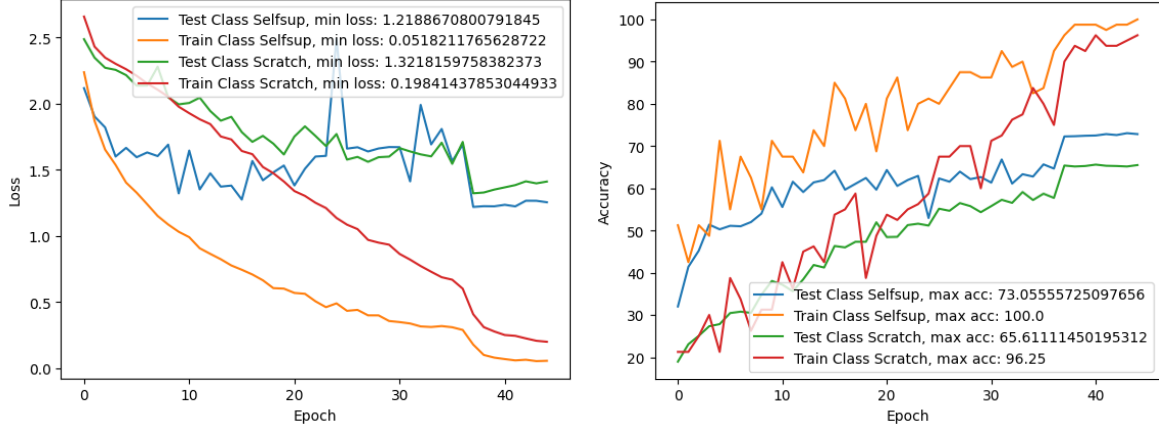


Figure 4: Comparison of training and validation loss (left) and accuracy (right) for classification when training from scratch vs. fine-tuning from pretext weights (45-epoch baseline configuration).

The full set of classification curves for all experimental variations (dataset size, learning rate, architecture, augmentation, and combined configuration) is included in Appendix B.

4.2 Discussion

Across all experiments, models initialized with self-supervised weights consistently train faster and more smoothly than those trained from scratch. As seen in Figure 4 and Appendix B (Fig. 15), fine-tuned models reach strong accuracy within the first few epochs, while scratch models need more time and show noisier validation curves. This difference is largest when using the reduced 15×480 dataset.

When using the full 50-class dataset (Appendix B, Fig. 16), the gap narrows. With more visual diversity, supervised learning becomes easier and less dependent on pretraining. Even then, fine-tuned models still converge earlier and exhibit smoother validation behaviour.

Regularization also plays a major role. Lower learning rates, stronger weight decay, and especially data augmentation all improve generalization (Appendix B, Figs. 19, 20). In these settings, scratch and fine-tuned models move closer together, not because pretraining becomes weaker, but because augmentation expands the effective training distribution.

Regarding model depth, ResNet-18 benefits more from pretraining (Appendix B, Fig. 18). The deeper architecture can exploit the richer features learned during the pretext task, but also overfits more easily when trained from scratch. This underscores that capacity alone is not enough: deeper models require either more data or stronger regularization.

The combined configuration (Appendix B, Fig. 21) shows the cumulative effect of all improvements. Both scratch and fine-tuned models perform well, but pretraining still provides faster convergence and more stable validation curves.

In summary, self-supervised pretraining helps most when data are limited or regularization is weak, but remains beneficial in all settings.

5 Model Visualizations

5.1 Representative t-SNE Visualizations

To better understand the structure of the learned feature space, we compute t-SNE embeddings for three models: pretext-only, fine-tuned (initialized from pretext weights), and scratch-trained. Figure 5 shows the representative t-SNE plots for the 45-epoch baseline configuration.

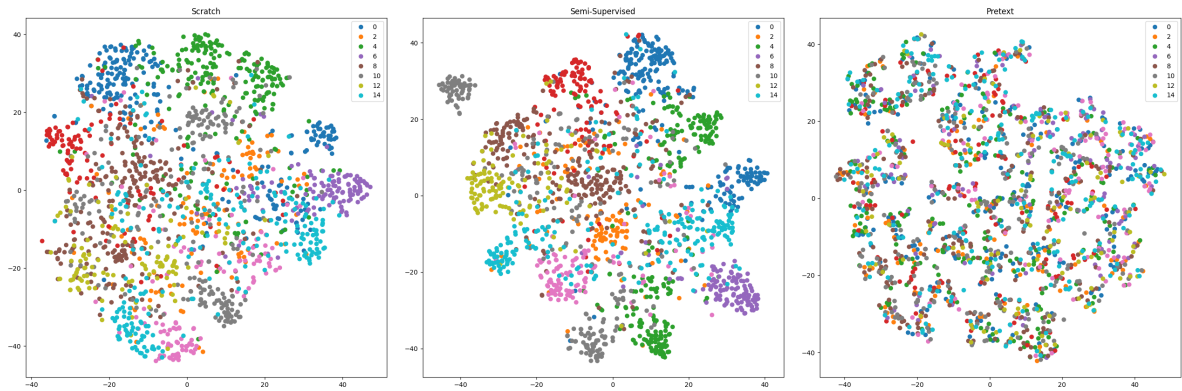


Figure 5: Representative t-SNE visualizations of the feature space for the scratch-trained classifier (left), the fine-tuned/self-supervised classifier (middle), and the pretext-only model (right), all trained for 45 epochs in the baseline configuration.

A comprehensive collection of t-SNE visualizations for all experimental settings is provided in Appendix C.

5.2 Discussion

The t-SNE visualizations clearly illustrate the differences between the three models. In the baseline case (Figure 5), the scratch-trained classifier produces diffuse clusters with noticeable overlap, reflecting the difficulty of training purely supervised models with limited data. The fine-tuned model, however, shows tighter and better separated clusters, meaning that the structure learned during the rotation task carries over to the supervised problem.

The pretext-only model naturally shows softer boundaries because it learns without labels. Even so, its embeddings are far from random: they form coherent regions that reflect shared visual patterns. This demonstrates that the pretext task captures useful low-level structure.

These trends appear consistently across all settings in Appendix C. With more classes (Appendix C, Fig. 24) or with stronger regularization and augmentation (Appendix C, Figs. 27 and 28), scratch-trained models improve noticeably. Still, the fine-tuned models almost always display cleaner separation and more stable cluster organization. The combined configuration (Appendix 29) shows this most clearly: both models look good, but the fine-tuned embeddings remain more structured.

Overall, the t-SNE results align strongly with the quantitative ones: pretraining consistently yields richer and more organized feature spaces, especially when labelled data are limited.

6 Conclusion

This work evaluated rotation-based self-supervised learning using ResNet-14 and ResNet-18 under a wide range of training conditions. Across all experiments, models initialized from the pretext task consistently trained faster, produced smoother validation behaviour, and learned more structured representations than models trained from scratch.

The results show that data diversity and augmentation play a central role. Increasing the number of classes or applying stronger augmentations greatly reduces overfitting and improves both pretext and downstream performance. In contrast, increasing model depth only provides clear benefits when paired with sufficient variability or regularization.

The t-SNE visualizations further support these findings: fine-tuned models consistently form clearer and more compact feature clusters, confirming that the representations learned during the pretext task transfer effectively to supervised classification.

Overall, rotation-based self-supervision proves to be a simple yet effective strategy, particularly when labelled data are limited. Although its relative advantage decreases as dataset diversity and regularization improve, it remains a reliable way to obtain better initial features and faster convergence across different training setups.

A Pretext Task: Full Results

This appendix contains the complete set of training and validation curves (loss and accuracy) for the pretext rotation prediction task under all experimental configurations described in the Methodology.

A.1 10-Epoch Sanity Check

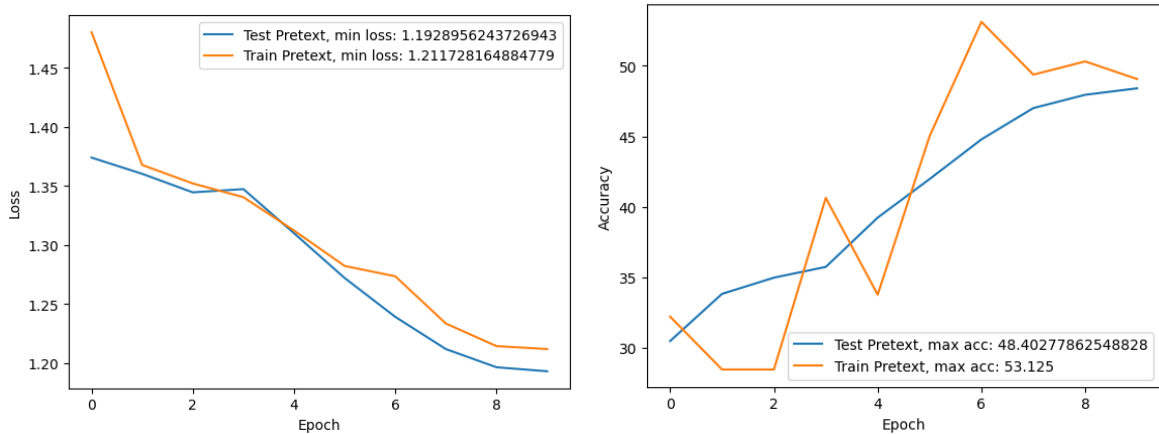


Figure 6: Pretext task (10-epoch sanity check): training and validation loss (left) and accuracy (right).

A.2 Baseline (45 Epochs, 15×480)

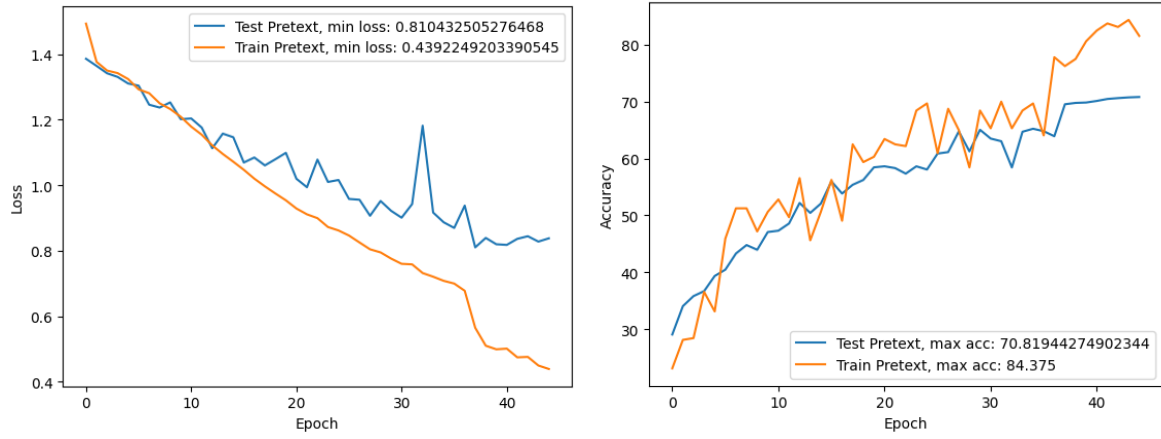


Figure 7: Pretext task baseline (45 epochs, ResNet-14, 15×480 dataset): training and validation loss (left) and accuracy (right).

A.3 Full Dataset (50×480)

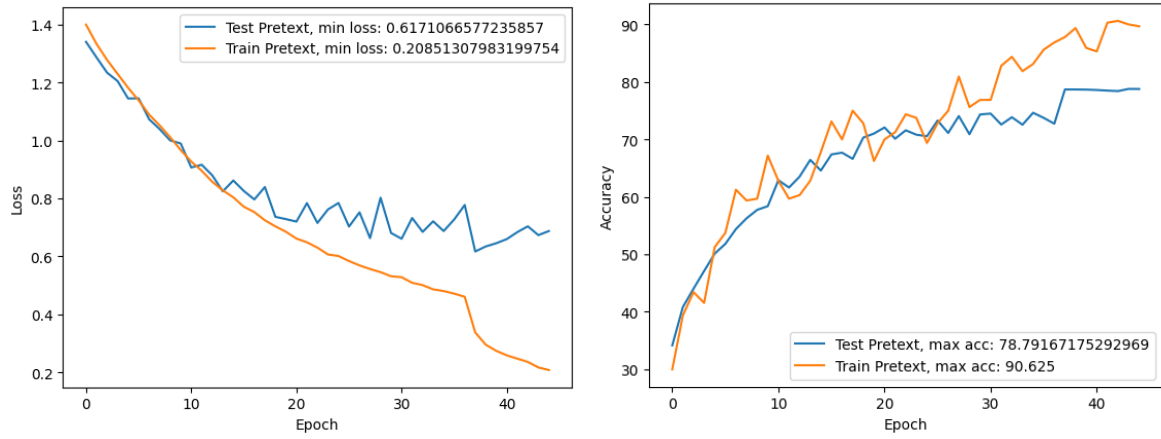


Figure 8: Pretext task with full dataset (50×480): training and validation loss (left) and accuracy (right).

A.4 Modified Learning Rate (0.01)

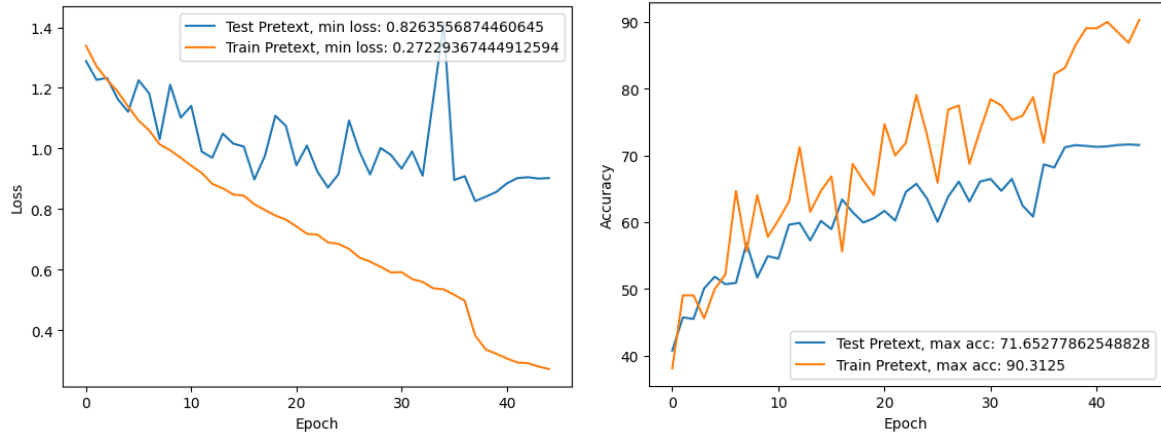


Figure 9: Pretext task with modified learning rate (0.01): training and validation loss (left) and accuracy (right).

A.5 ResNet-18 Backbone

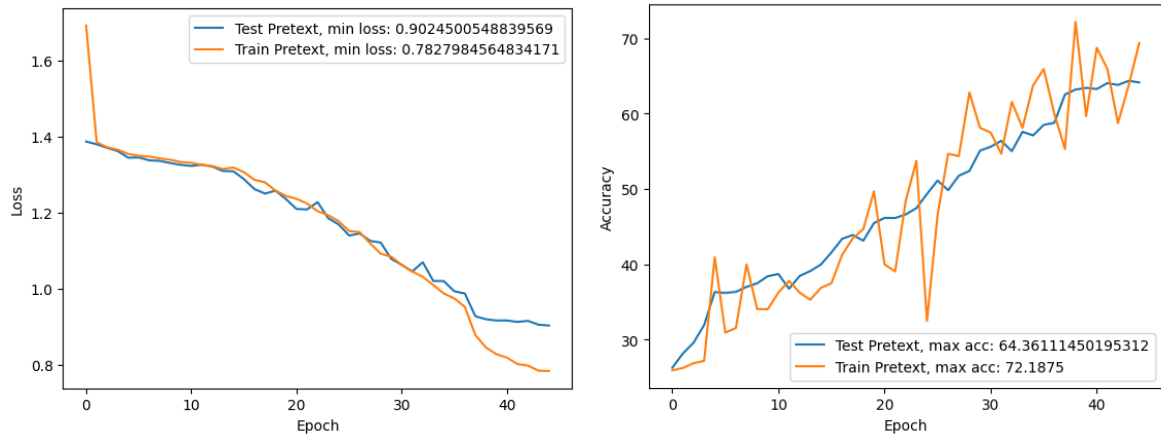


Figure 10: Pretext task with ResNet-18 backbone: training and validation loss (left) and accuracy (right).

A.6 Data Augmentation (ResNet-14)

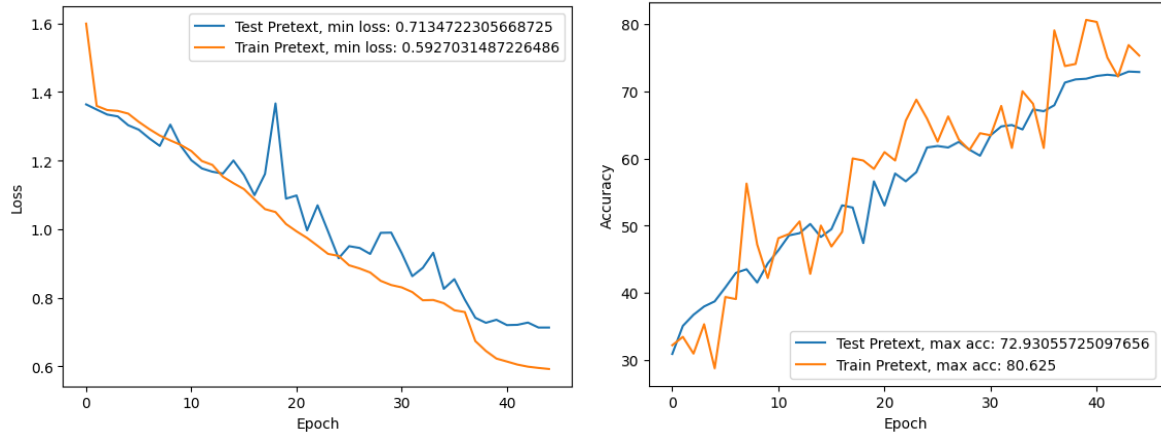


Figure 11: Pretext task with data augmentation (ResNet-14): training and validation loss (left) and accuracy (right).

A.7 Data Augmentation (ResNet-18)

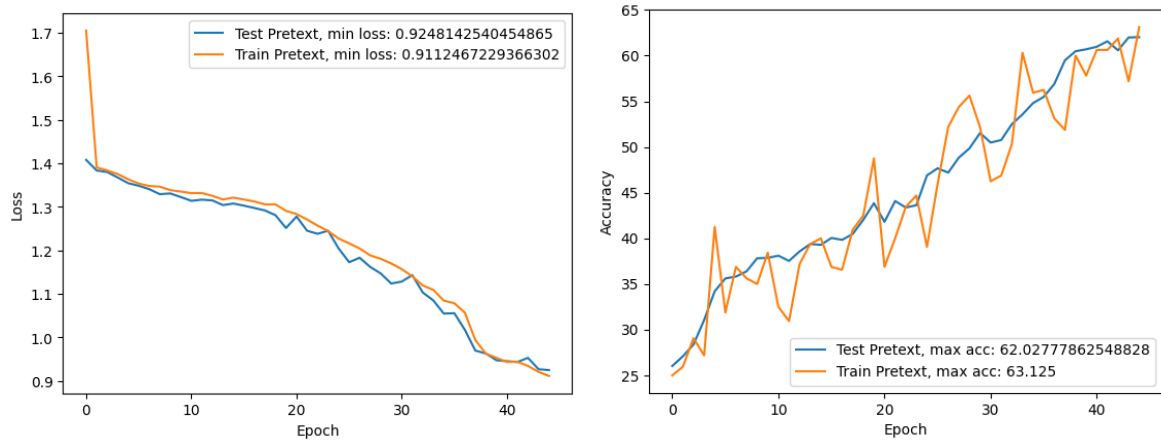


Figure 12: Pretext task with data augmentation (ResNet-18): training and validation loss (left) and accuracy (right).

A.8 Combined Configuration

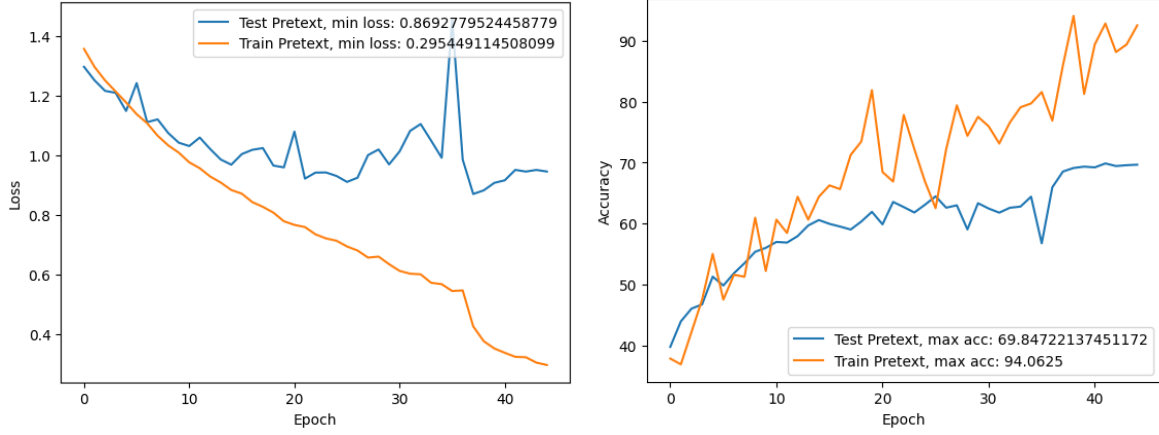


Figure 13: Pretext task with combined configuration (ResNet-18, reduced learning rate, increased weight decay, and data augmentation): training and validation loss (left) and accuracy (right).

B Classification Task: Full Results

This appendix reports the complete set of training and validation curves for the supervised classification task, comparing scratch training and initialization from self-supervised pretext weights.

B.1 10-Epoch Sanity Check

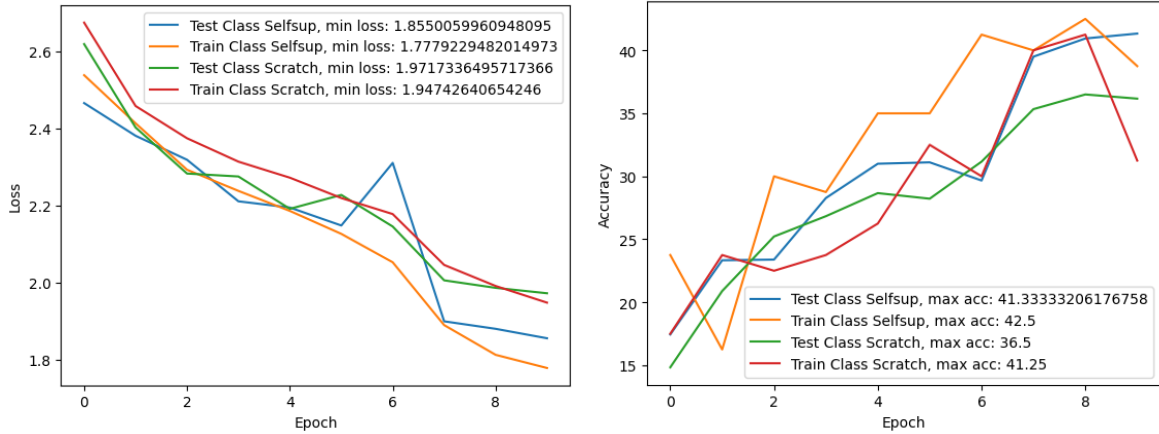


Figure 14: Classification task (10-epoch sanity check): training and validation loss (left) and accuracy (right) for scratch vs. self-supervised initialization.

B.2 Baseline (45 Epochs, 15×480)

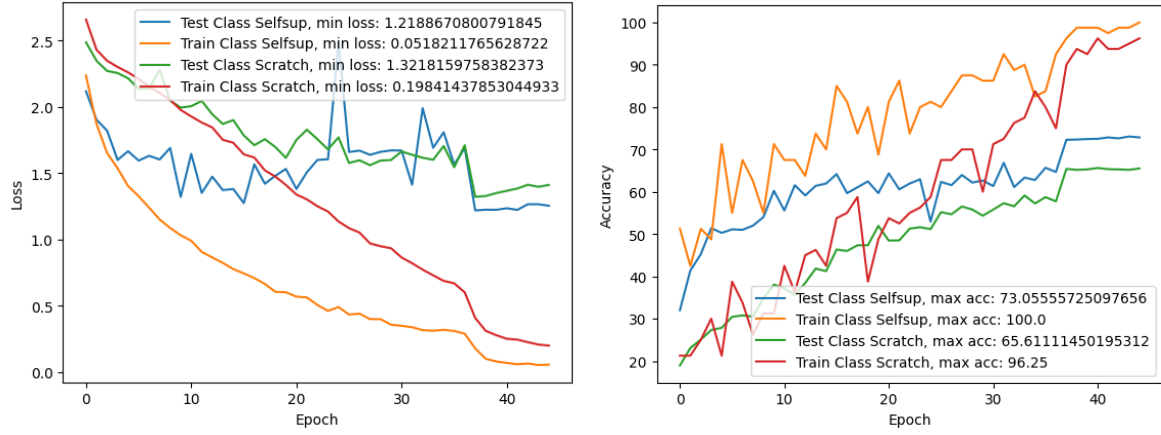


Figure 15: Classification task baseline (45 epochs, 15×480): training and validation loss (left) and accuracy (right) for scratch vs. self-supervised initialization.

B.3 Full Dataset (50×480)

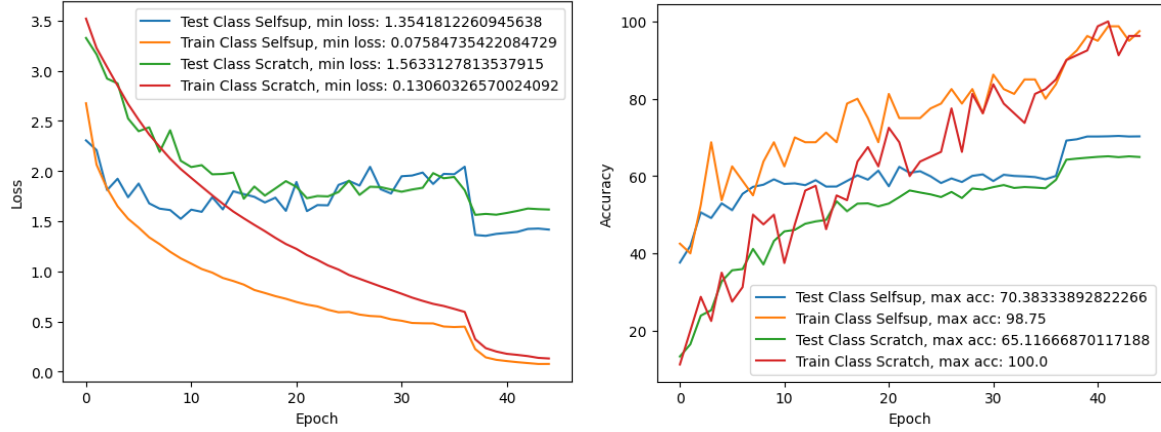


Figure 16: Classification task with full dataset (50×480): training and validation loss (left) and accuracy (right) for scratch vs. self-supervised initialization.

B.4 Modified Learning Rate (0.01)

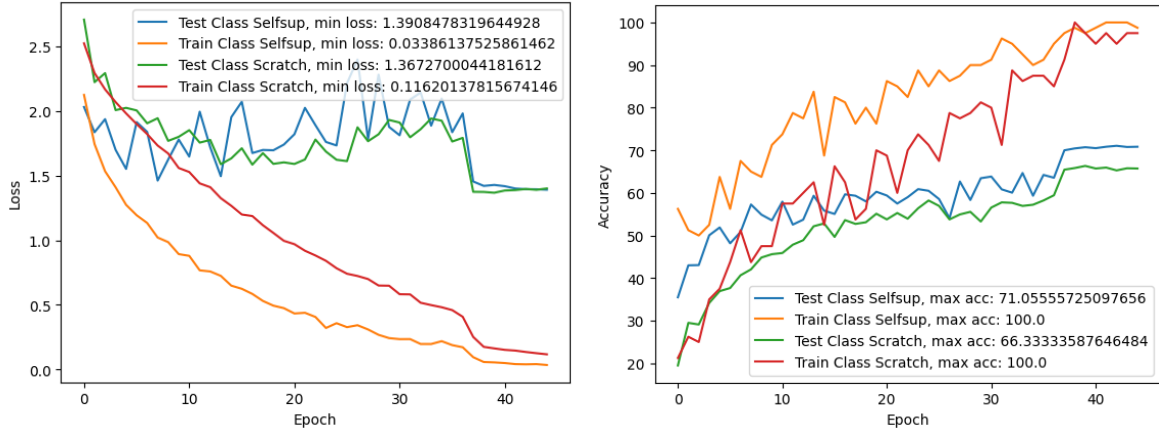


Figure 17: Classification task with modified learning rate (0.01): training and validation loss (left) and accuracy (right) for scratch vs. self-supervised initialization.

B.5 ResNet-18 Backbone

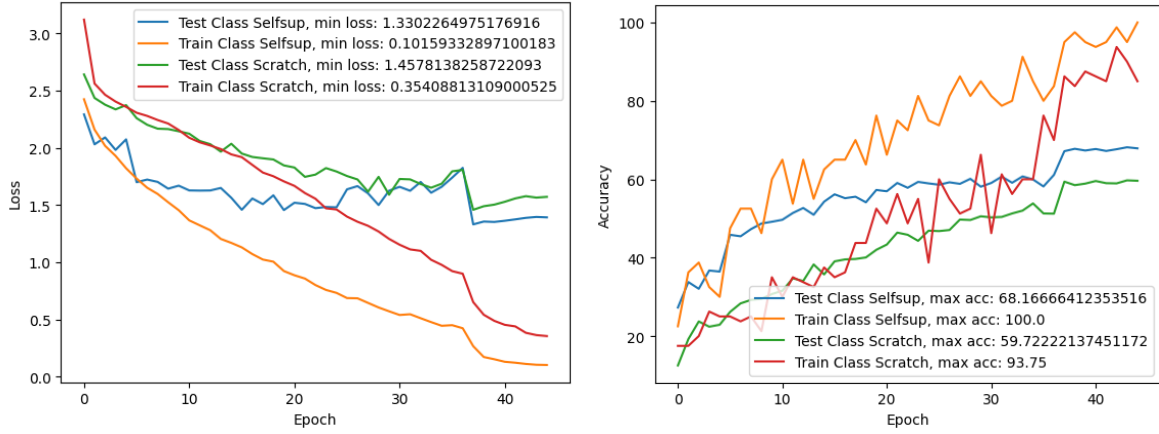


Figure 18: Classification task with ResNet-18 backbone: training and validation loss (left) and accuracy (right) for scratch vs. self-supervised initialization.

B.6 Data Augmentation (ResNet-14)

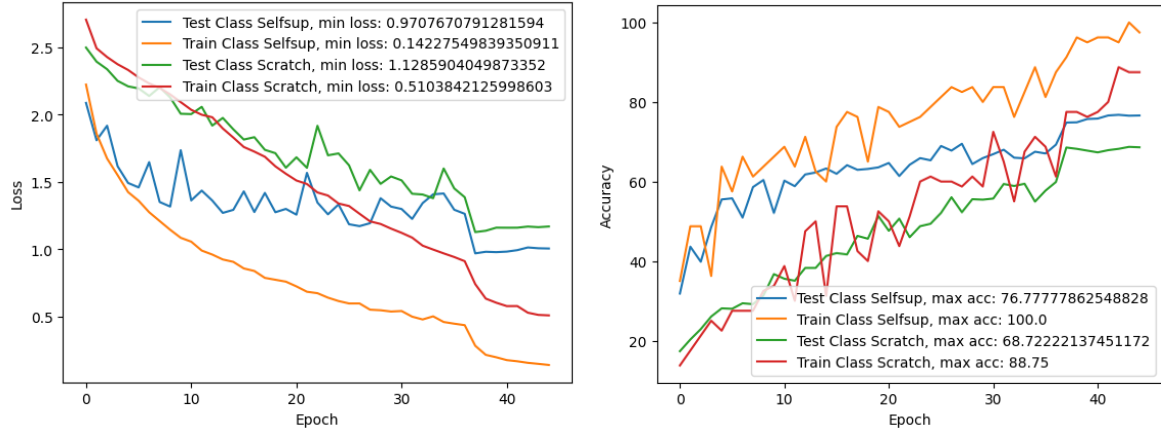


Figure 19: Classification task with data augmentation (ResNet-14): training and validation loss (left) and accuracy (right) for scratch vs. self-supervised initialization.

B.7 Data Augmentation (ResNet-18)

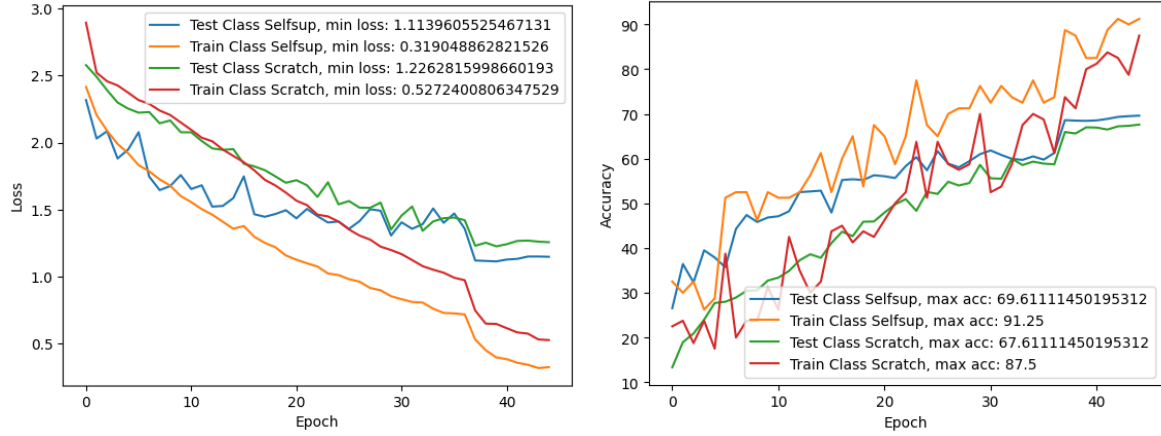


Figure 20: Classification task with data augmentation (ResNet-18): training and validation loss (left) and accuracy (right) for scratch vs. self-supervised initialization.

B.8 Combined Configuration

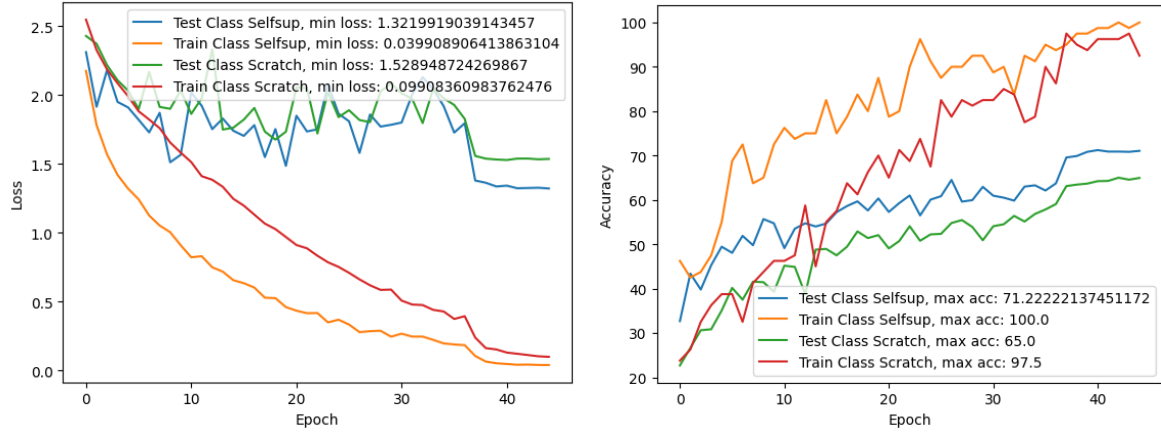


Figure 21: Classification task with combined configuration (ResNet-18, reduced learning rate, increased weight decay, and data augmentation): training and validation loss (left) and accuracy (right) for scratch vs. self-supervised initialization.

C t-SNE Visualizations: Full Results

This appendix includes the complete set of t-SNE visualizations for all configurations. Each figure shows the feature space embeddings for the scratch-trained classifier, the fine-tuned/self-supervised classifier, and the pretext-only model.

C.1 10-Epoch Sanity Check

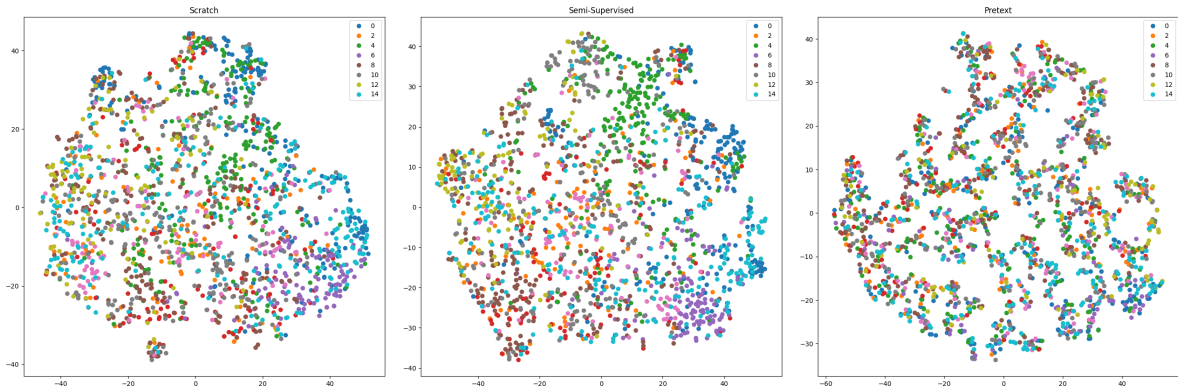


Figure 22: t-SNE visualizations (10-epoch sanity check): scratch (left), self-supervised/fine-tuned (middle), pretext-only (right).

C.2 Baseline (45 Epochs, 15×480)

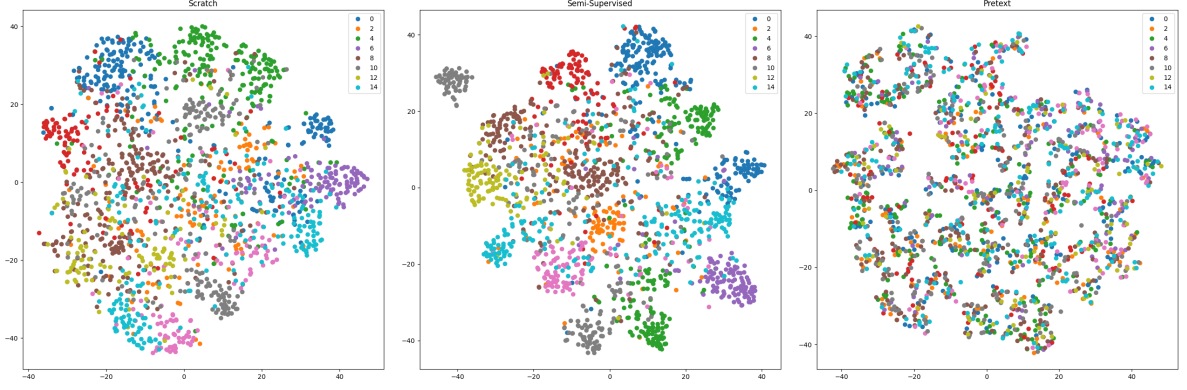


Figure 23: t-SNE visualizations baseline (45 epochs, 15×480): scratch (left), self-supervised/fine-tuned (middle), pretext-only (right).

C.3 Full Dataset (50×480)

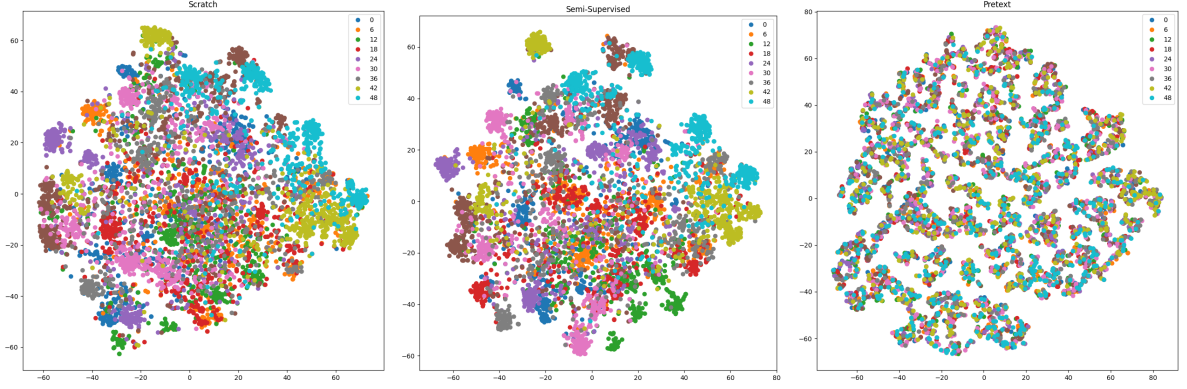


Figure 24: t-SNE visualizations with full dataset (50×480): scratch (left), self-supervised/fine-tuned (middle), pretext-only (right).

C.4 Modified Learning Rate (0.01)

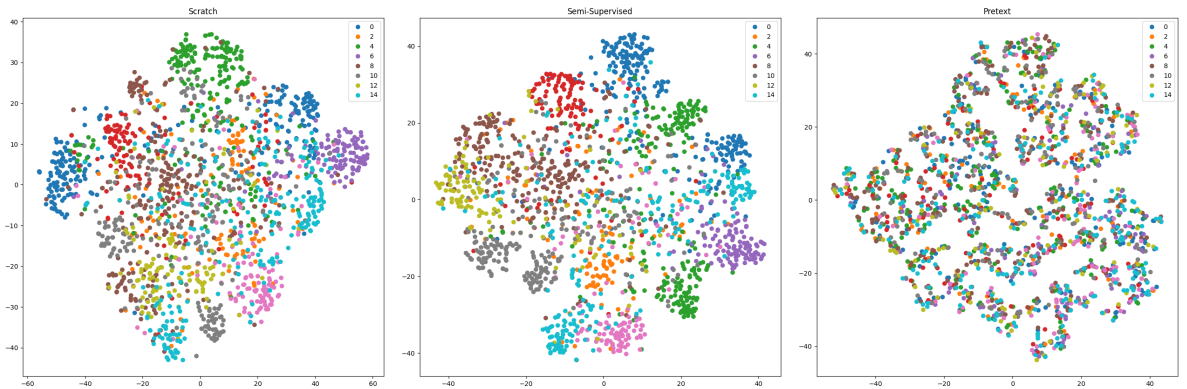


Figure 25: t-SNE visualizations with modified learning rate (0.01): scratch (left), self-supervised/fine-tuned (middle), pretext-only (right).

C.5 ResNet-18 Backbone

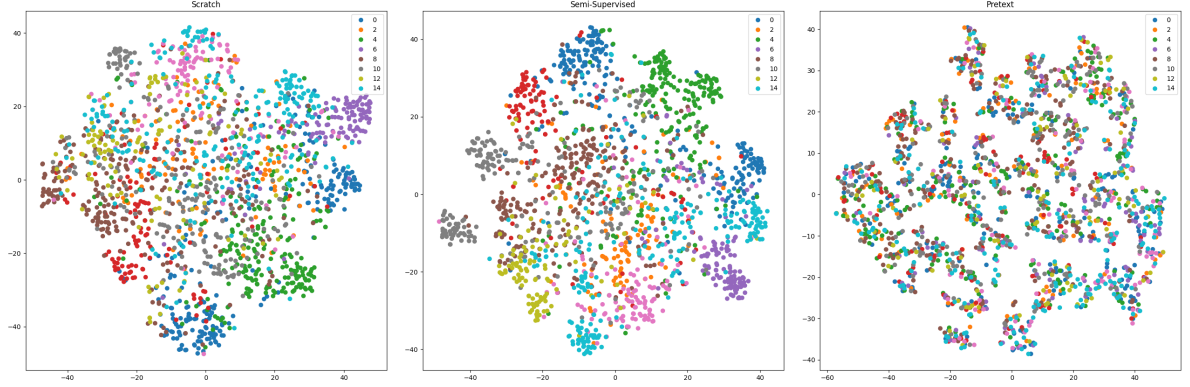


Figure 26: t-SNE visualizations with ResNet-18 backbone: scratch (left), self-supervised/fine-tuned (middle), pretext-only (right).

C.6 Data Augmentation (ResNet-14)

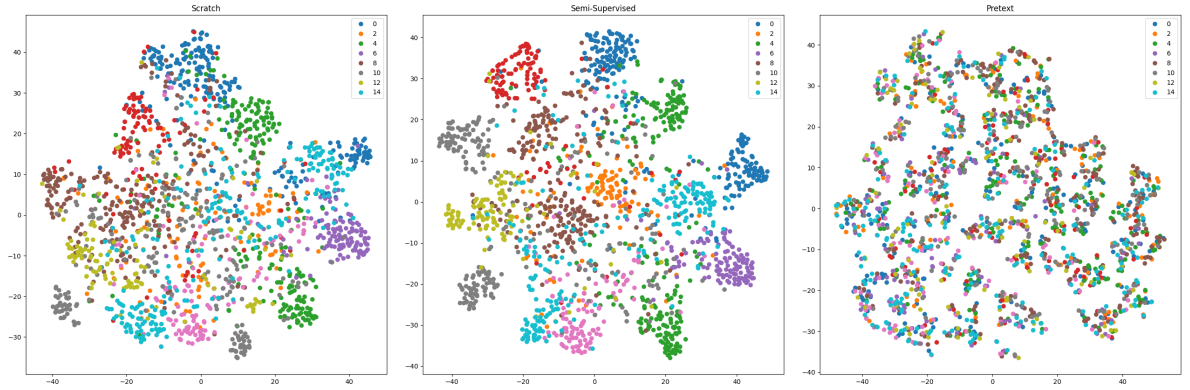


Figure 27: t-SNE visualizations with data augmentation (ResNet-14): scratch (left), self-supervised/fine-tuned (middle), pretext-only (right).

C.7 Data Augmentation (ResNet-18)

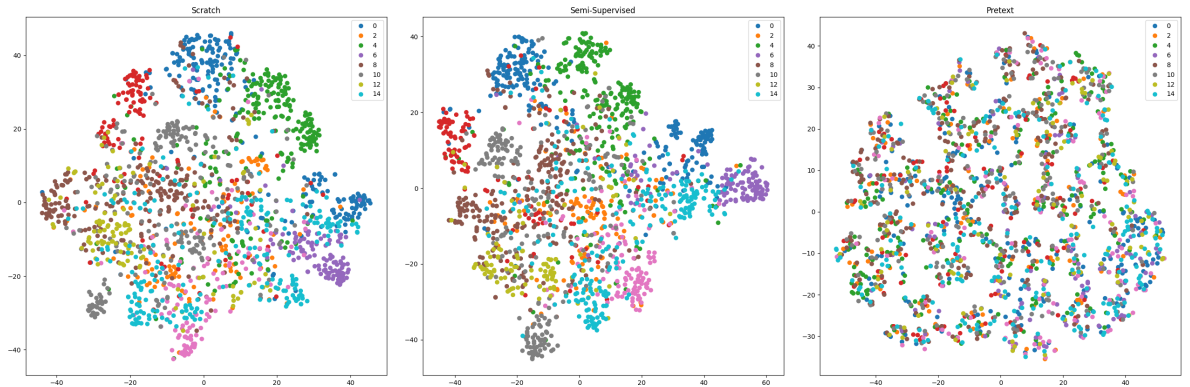


Figure 28: t-SNE visualizations with data augmentation (ResNet-18): scratch (left), self-supervised/fine-tuned (middle), pretext-only (right).

C.8 Combined Configuration

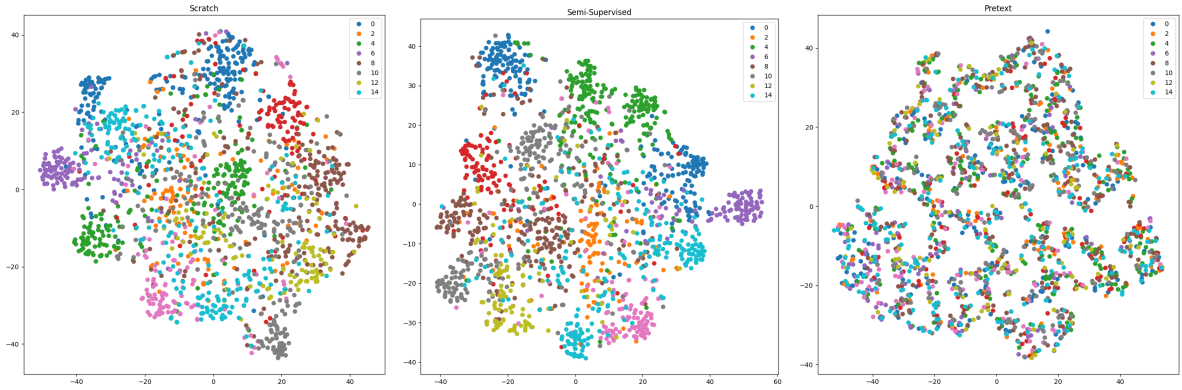


Figure 29: t-SNE visualizations with combined configuration (ResNet-18, reduced learning rate, increased weight decay, and data augmentation): scratch (left), self-supervised/fine-tuned (middle), pretext-only (right).

References

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.